

PICK A CHART — ALL 36 TYPES, BY INTENT

COMPARE

<code>pv_bar()</code>	magnitudes by category; stacks by value or to 100%
<code>pv_lollipop()</code>	a ranking with less ink than bars
<code>pv_slope()</code>	two moments, one line per group; the slope is the change
<code>pv_dumbbell()</code>	two values per category, the gap on every row
<code>pv_waterfall()</code>	signed contributions building to a total
<code>pv_bullet()</code>	measures against their targets, in compact rows
<code>pv_heatmap()</code>	one value across two category axes

CHANGE OVER TIME

<code>pv_line()</code>	a measure over time, a few series
<code>pv_area()</code>	totals over time: stacked, percent or stream
<code>pv_horizon()</code>	dozens of series, folded into compact ribbons
<code>pv_bump()</code>	ranks trading places, period by period
<code>pv_race()</code>	the animated ranking, one frame per period
<code>pv_calendar()</code>	a daily value, laid out year by year

DISTRIBUTION

<code>pv_histogram()</code>	the shape of one numeric variable
<code>pv_boxplot()</code>	five-number summaries across groups
<code>pv_violin()</code>	full densities, group against group
<code>pv_ridgeline()</code>	many densities stacked in rows
<code>pv_beeswarm()</code>	every point visible, none overlapping

COMPOSITION

<code>pv_donut()</code>	a few shares of one whole
<code>pv_waffle()</code>	shares as countable unit squares
<code>pv_treemap()</code>	many parts sized by value, one or two levels
<code>pv_sunburst()</code>	a hierarchy in rings from the root; zoomable
<code>pv_icicle()</code>	the same hierarchy as columns, labels legible
<code>pv_pack()</code>	a hierarchy as nested circles; zoomable

RELATIONSHIPS

<code>pv_scatter()</code>	two numerics; contours or hex bins when crowded
<code>pv_pairs()</code>	every numeric pair at once, as a matrix
<code>pv_parallel()</code>	many numeric columns, brushable
<code>pv_sankey()</code>	flows between stages, width carrying volume
<code>pv_chord()</code>	flows around one closed set, from a matrix
<code>pv_arc()</code>	a network on one line, crossings minimised
<code>pv_force()</code>	a network left free to find its clusters

MAPS

<code>pv_choropleth()</code>	a value per region, shaded on its map
<code>pv_bubble_map()</code>	sized points at real coordinates
<code>pv_flow_map()</code>	movement between places, as tapered bands

SPECIAL

<code>pv_table()</code>	the numbers themselves: sortable, in-cell bars, sparklines
<code>pv_dendrogram()</code>	an hclust tree, cut into k groups

THE SHARED GRAMMAR

EVERY CHART

```
pv_bar(data, x, y, series = NULL,
        title, subtitle, source,
        mode = "auto")
```

All 36 constructors share the title block, the source line and `mode: "auto"` follows the reader's light or dark side, `"light"` and `"dark"` pin it. Wherever a chart decides for itself — orientation, legends, value labels — the argument takes `TRUE`, `FALSE` or `"auto"`.

Degenerate data fails at once with a plain R message, never a broken chart.

PIPE-ABLE LAYERS

Every modifier takes a chart and hands it back:
`w |> pv_trend()` |> `pv_annotate(...)`
`pv_annotate(...)` — layers `pv_hline()`, `pv_vline()`, `pv_band()` and `pv_note()` onto the plot, in data units
`pv_trend()` — a fitted trend and its ribbon: `"loess"` or `"lm"`
`pv_facet(by)` — small multiples on one variable, scales shared
`pv_textures()` — hatches each series, so print and photocopies keep identity
`pv_static()` — no animation, tooltips or controls
`pv_downloads(FALSE)` — hides the hover download button
`pv_link(sd)` — crosstalk selection linking across charts
`pv_alt("...")` — your own words for screen readers; `pv_alt_text(w)` reads what a chart would say

COMPOSE & STATISTICS

`pv_board(a, b)` — finished charts composed into one page; `pv_save()` takes the board whole
`pv_decompose(d, x, y)` — trend / seasonal / remainder panels — `stats::stl()` or classical
`w |> pv_forecast(h)` — a fan of model-based intervals: Holt-Winters, small ARIMA, naive
`w |> pv_changepoints()` — marks sustained level shifts — binary segmentation + BIC

PAPER, DECKS & DATA

ONTO PAPER

`pv_save(w, "chart.png")` — the extension picks the format:

- `.png` crisp 2× raster, for Word and slides
- `.svg` standalone vector, font embedded
- `.pdf` a true vector print — made for LaTeX
- `.gif` the bar-chart race, a frame-exact loop
- `.html` still interactive; needs no Chrome

`pv_deck(charts, "review.pptx")` — a list of charts to PowerPoint, one 2× capture per slide

KNITTING TO WORD & PDF

```
knitr::opts_chunk$set(
  screenshot.force = FALSE)
```

The one setup line: in output that cannot run d3, a chunk's chart becomes a print-quality figure at the chunk's size and dpi.

DATA IN

`pv_read("f.csv")` — one reader for `.csv`, `.sqlite`, `.sas7bdat` and `.xpt`
`pv_search_opendata(q)` — search the federal catalogue; `pv_fetch_opendata()` fetches what it lists
`pv_fetch_bfs(id)` — a stats.swiss SDMX dataflow, tidied
`pv_fetch_lustat(id)` — a LUSTAT Statistik Luzern table
`pv_fetch_map("municipalities")` — the one boundary layer too big to bundle
Each fetch states its source and licence; downloads cache once (`pv_cache_status()`).

THEME & LOCALE

`pv_set_theme(pv_theme_paper())` — the print-first paper theme; `pv_reset_theme()` undoes it
`pv_locale("de-CH")` — numbers and dates the Swiss way (10'000, März); `fr/it/en-CH` too; `NULL` resets

BUNDLED DATA

17 datasets on board: `pv_sales`, `pv_network` and `pv_flows` are simulated; the rest is real Swiss open data — cities, commuters, fiscal equalization, elections, weather, electricity, tourism — plus the maps' boundary layers.